

# [ACJC/2024/Q2]

## 2 Name your Jupyter Notebook as:

TASK2\_<your name>\_<centre number>\_<index number>.ipynb

The task is to create a game where players will try to locate "mines" which are hidden at random locations in field which is a rectangular grid. The player's task is to determine where the mines are.

On each turn, the player selects one square. If the player selects a square with a mine, it blows up and the game is over. Otherwise, the player is told how many mines there are in the eight squares surrounding the selected square, and the player's next turn begins. If the player manages to select all the squares without mines before blowing up any mines, the player wins.

For example, the diagram below shows a 3 x 4 grid. There are two mines hidden at (0, 0) (the 0<sup>th</sup> row and the 0<sup>th</sup> column, i.e. the top left corner) and (1, 2) (the 1<sup>st</sup> row and the 2<sup>nd</sup> column), as indicated by the X's.

```
X...
..X.
....
```

Since the player does not know where the mines are, the field is displayed to the player as follows.

```
....
....
....
```

If the player selects (1, 1) on the first turn, the player is told that there are 2 mines in the eight squares surrounding the (1, 1). The field is then displayed to the player as follows.

```
....
.2..
....
```

If the player selects (0, 2) on the second turn, the player is told that there is 1 mine in the five squares surrounding (0, 2) .

```
..1.
.2..
....
```

If the player selects (0, 0) on the third turn, the mine at (0, 0) blows up and the game ends.

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 2.1
        Program code
        Output:
```

### Task 2.1

The player sets up the game by giving the dimensions of the field and the number of mines.

Write code to:

- ask the player for the height and width of the field;
- ask the player for the number of mines;
- validate that the number of mines is larger than zero and smaller than the size of the field;
- create the field by putting the mines at random locations in the field.

There should only be at most one mine at each location in the field.

[7]

# [ACJC/2024/Q2 Cont']

## Task 2.2

The player can now play the game as described above.

Write code to:

- display the field to the player as described above;
- ask the player for the row and column number of the location to try;
- end the game if the player's location has a mine, or, otherwise, reveal the number of mines in the squares surrounding it;
- repeat the above steps until the player selects a mine, or has revealed all squares without mines.

Your program should print appropriate messages to the player when the game ends because the player selects a mine, or because the player wins. [11]

Save your Jupyter Notebook for Task 2.